

Information and Communication Technology (CS 407)

M.C.A 1st Semester

Autumn Semester Mid-Term Examination 2017

Computer Science and Engineering

TU/CSE

Time Duration: 75 mins

Total Marks:40

Note: Please write your answers precisely and to the point.

1. Write *precise, proper* and *relevant justification* for the following incorrect statements.

2 X 9 = 18 Marks

- ~~F~~ (a) The technology shift from single core processor to multi-core processors is mainly because of the cost to have one single large core processor. ~~F~~
- ~~T~~ (b) Processors based on *Demand based Switching* technology are designed so that the core with most number of processes running within it gets the higher priority to execute its processes than the other cores in multi-core processors.
- ~~F~~ (c) Primary storage units such as RAM have lower cost per bit of storage as compared to secondary storage units such as hard drive.
- ~~T~~ (d) Character addressable computer are faster than the word addressable computer.
- ~~F~~ (e) Bus clock cycle is mainly related to speed of the memory bus connecting the physical memory and the north bridge.
- ~~F~~ (f) RISC processors claims higher speed gain mainly and only because of the use of fixed-length instructions.
- ~~T~~ (g) The number of bits supported by a processor is mainly decided by the number of bits for a word in the cache memory.
- ~~T~~ (i) The assignment of a predicate register to an instruction in EPIC processors is mainly to check during run time that the instruction loaded in the instruction register is valid or not.
- ~~T~~ (i) DDR SDRAM is faster than SRAM or DRAM because the transmission of data is based only the synchronization to the CPU clock cycle.

2. A system you've purchased is of 32-bit processor with a RAM 4 GB. Do you think that only updating the RAM to 8GB can increase the overall performance of your system expecting that more RAM size means more number of applications you can run. Give reasons.

3 Marks

3. Why we cannot increase the size of SRAM as much as we prefer in cache memory though we know that SRAM is faster than DRAM ?

2 Marks

4. What do you mean by *Locality of Reference* ? Explain with a programming example clearly.

2+3 = 5 Marks

5. Write pointwise why do we have UNICODE system? Write pointwise how UTF-8 differs from UTF-16 ?

2 X 3 = 6 Marks

6. Explain with good scenario or example how block size breaks the principle of *Locality of Reference*? Write about the two extremes of *Write Policy* in cache.

4 + 2 = 6 Marks

*****End of Paper-Best of Luck*****